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(54) TECHNIQUES FOR AUTOMATED BEAM ADJUSTMENT FOR PORTAL DIRECTIONALITY

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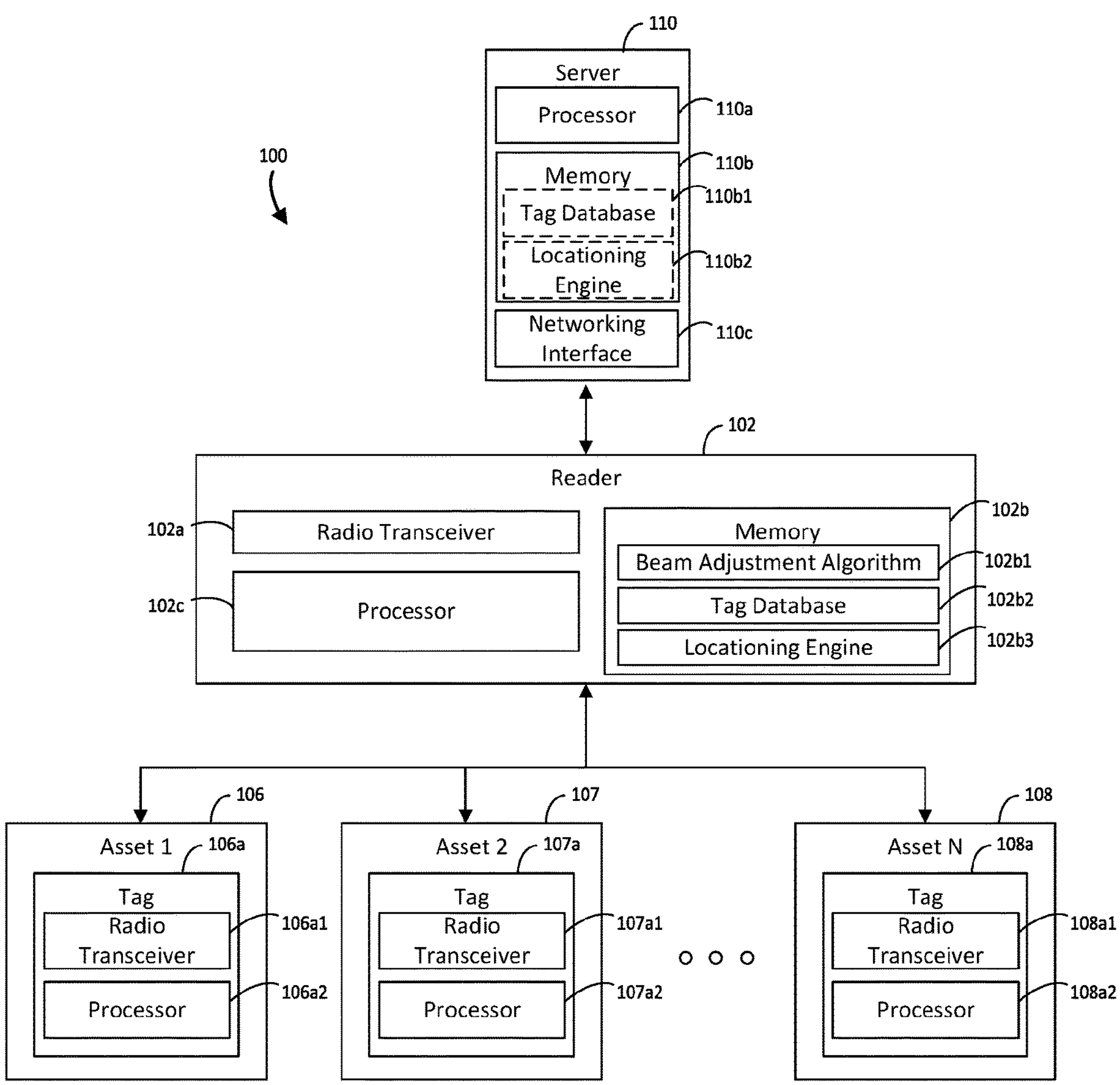
(57) **ABSTRACT**

Techniques for automated beam adjustment for portal directionality are disclosed herein. An example apparatus includes: a transceiver configured to emit a plurality of signal beams that each have a respective coverage area and are oriented in a respective direction, one or more processors, and one or more memories communicatively coupled to the one or more processors storing a beam adjustment algorithm. The example apparatus may include instructions that, when executed cause the assembly to: transmit a first signal beam to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction; determine, by the beam adjustment algorithm, that a first tag threshold is violated; and adjust, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction.



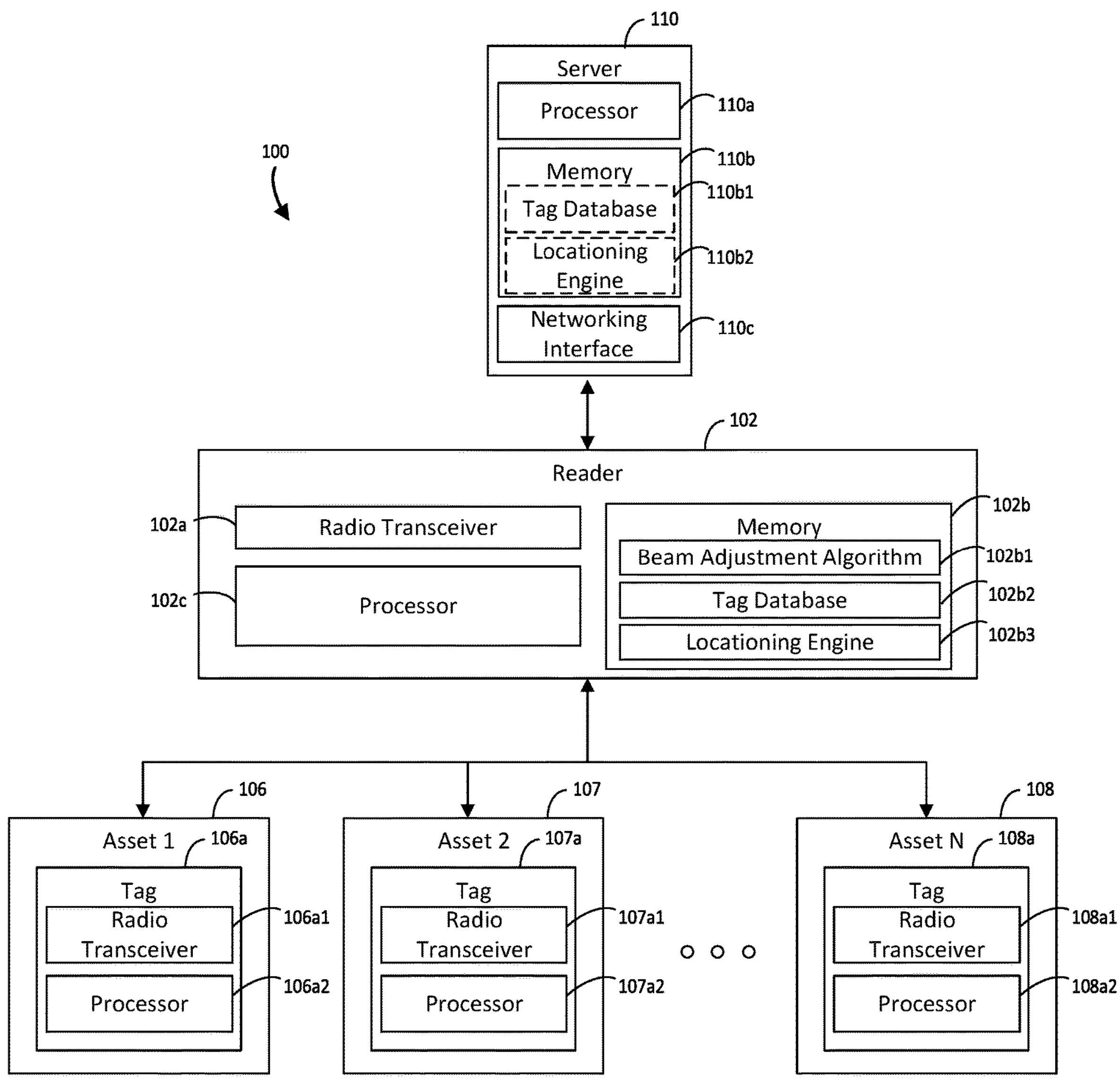
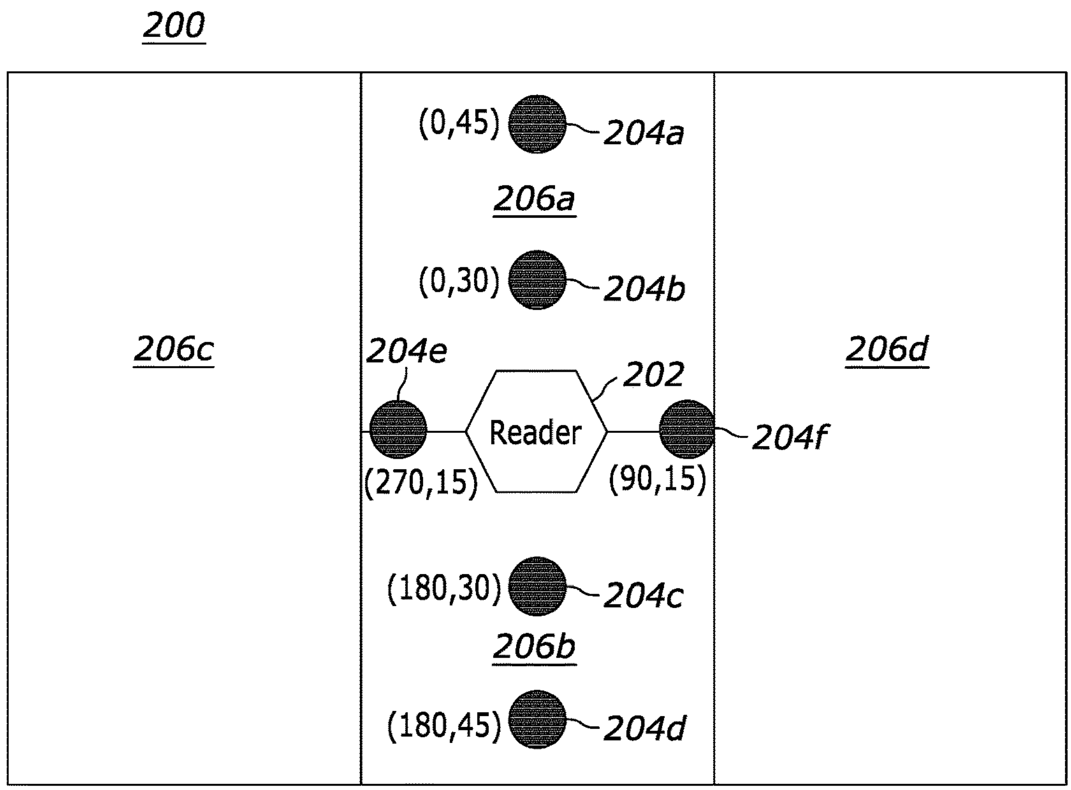

110
Server
Processor
110a
Memory
110b
Tag Database
110b1
Locationing Engine
110b2
Networking Interface
110c
100
102
Reader
Radio Transceiver
102a
Processor
102c
Memory
102b
Beam Adjustment Algorithm
102b1
Tag Database
102b2
Locationing Engine
102b3
106
Asset 1
106a
Tag
Radio Transceiver
106a1
Processor
106a2
107
Asset 2
107a
Tag
Radio Transceiver
107a1
Processor
107a2
108
Asset N
108a
Tag
Radio Transceiver
108a1
Processor
108a2

**FIG. 1**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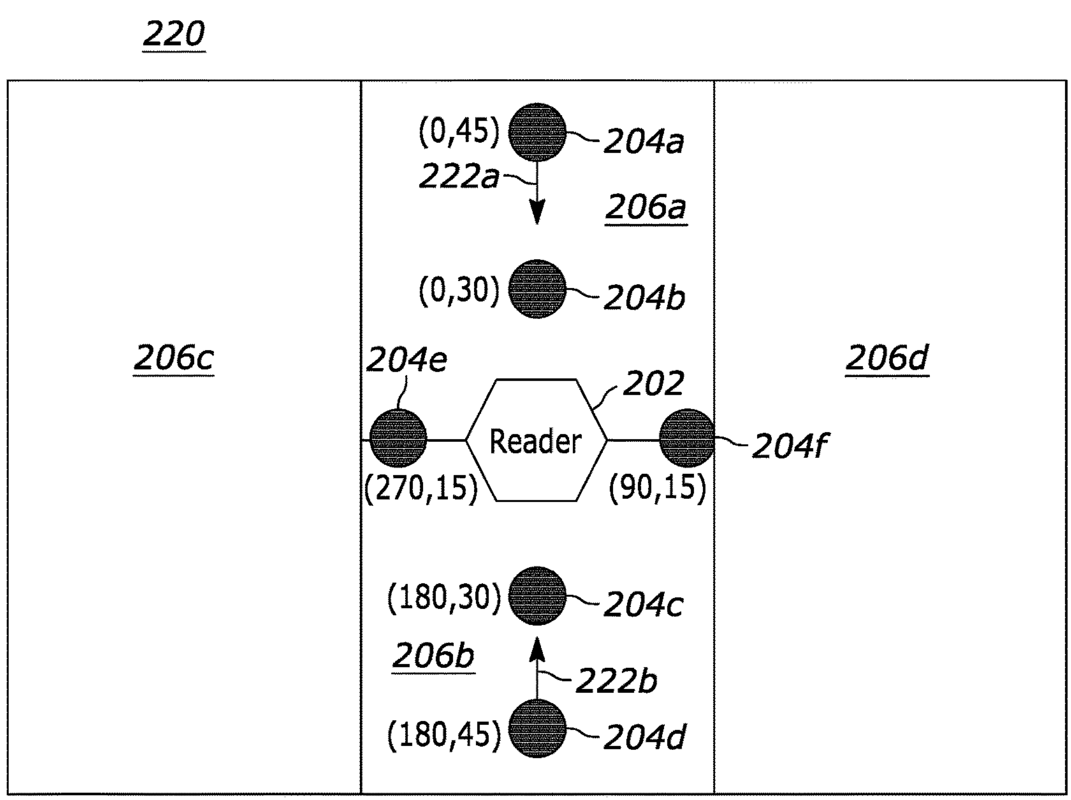
204a
206a
(0,30)
204b
206c
204e
202
206d
Reader
204f
(270,15)
(90,15)
(180,30)
204c
206b
(180,45)
204d

FIG. 2A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222a
206a
(0,30)
204b
206c
204e
202
206d
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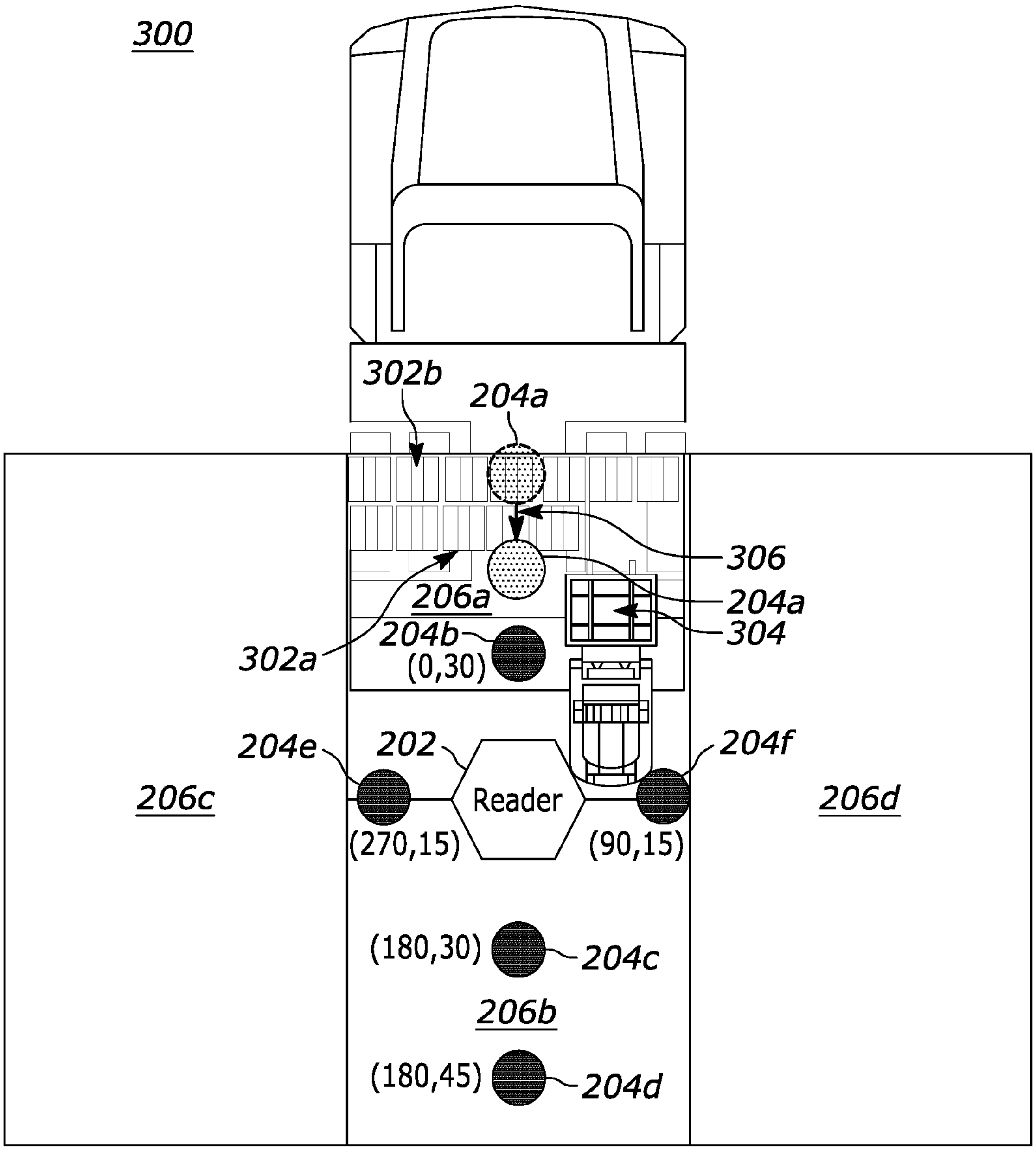
(90,15)
(180,30)
204c
206b
222b
(180,45)
204d

FIG. 2B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302b
204a
306
204a
304
206a
302a
204b
(0,30)
204e
202
204f
206c
Reader
206d
(270,15)
(90,15)
(180,30)
204c
206b
(180,45)
204d

FIG. 3A

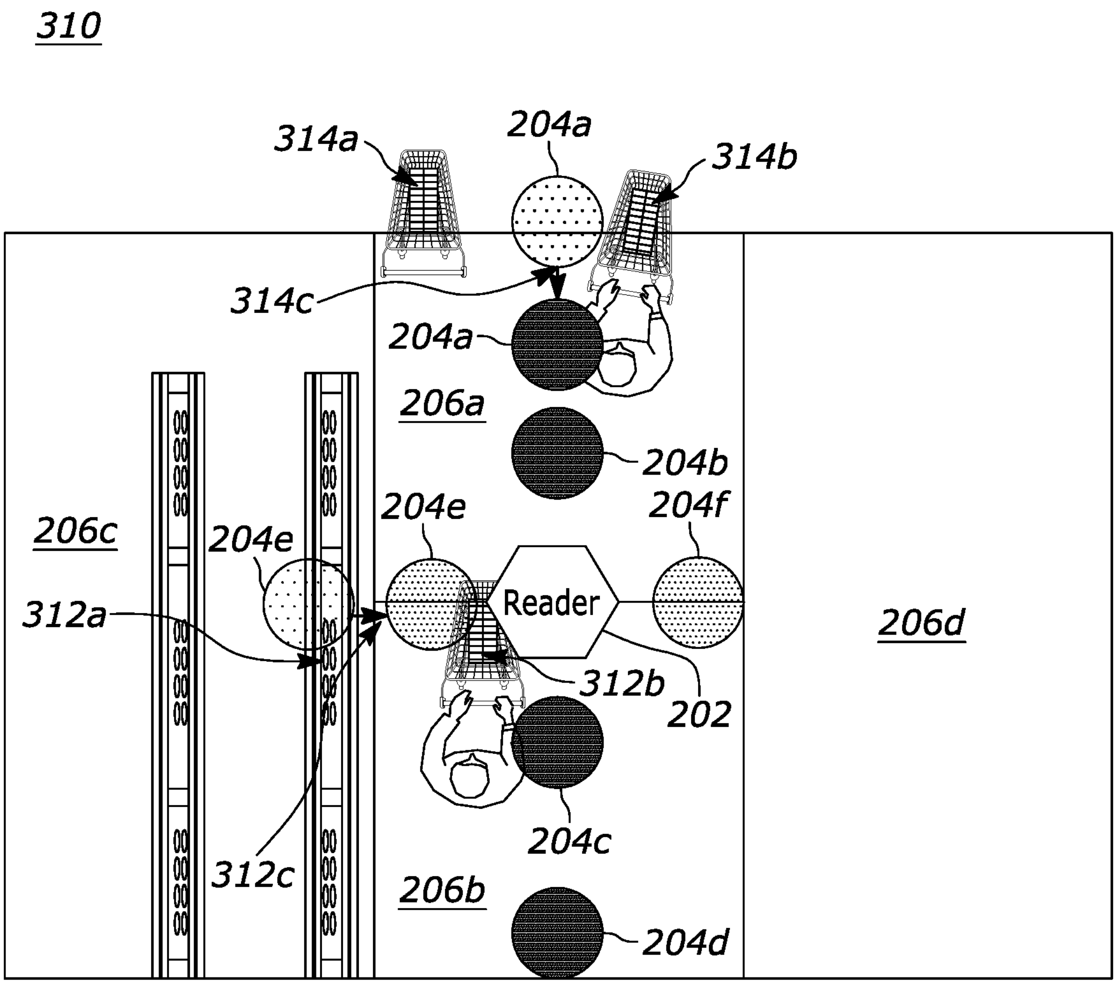
310
314a
204a
314b
314c
204a
206a
204b
206c
204e
204e
204f
312a
Reader
206d
312b
202
204c
312c
206b
204d

FIG. 3B

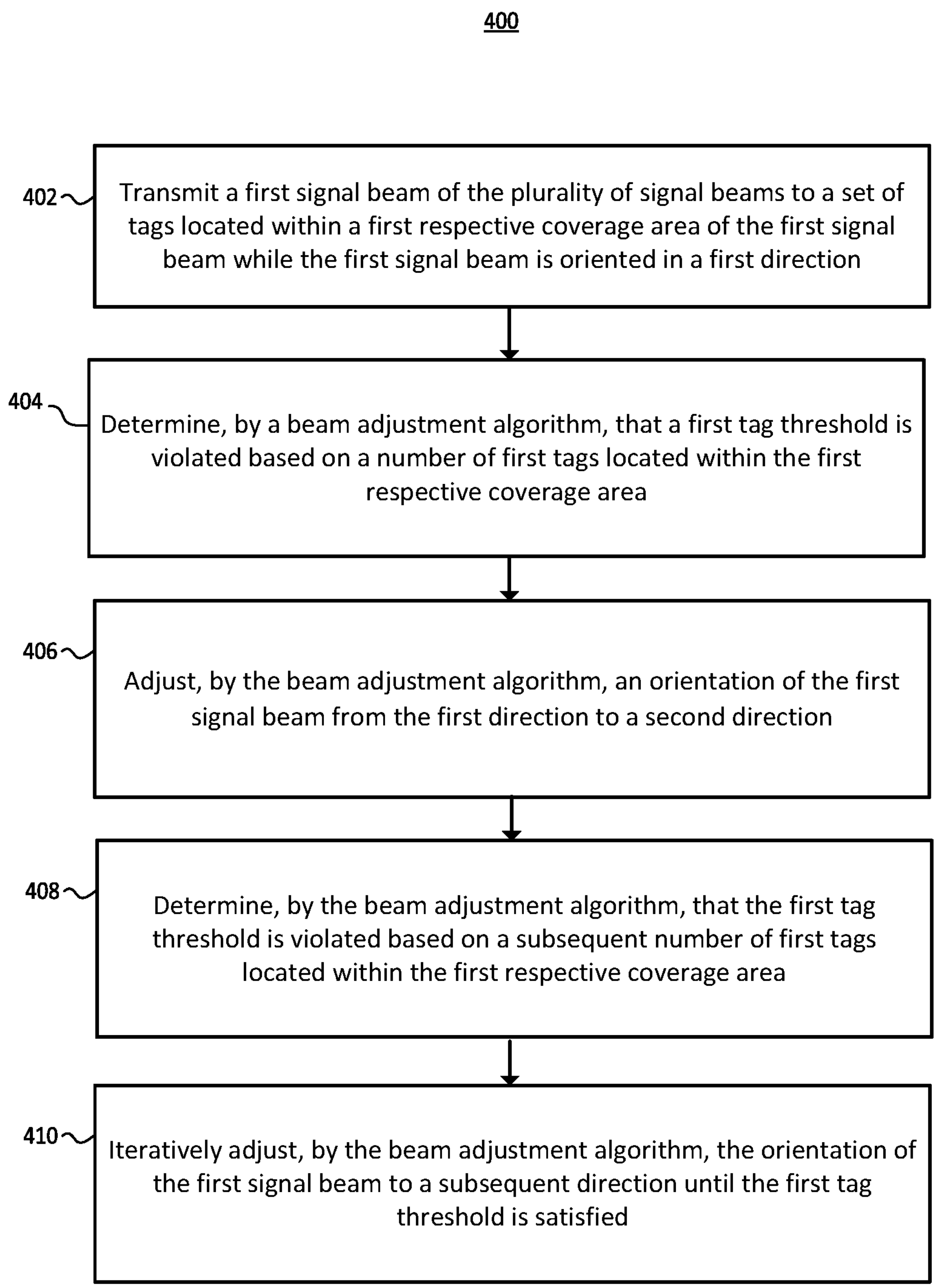
400
402
Transmit a first signal beam of the plurality of signal beams to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction
404
Determine, by a beam adjustment algorithm, that a first tag threshold is violated based on a number of first tags located within the first respective coverage area
406
Adjust, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction
408
Determine, by the beam adjustment algorithm, that the first tag threshold is violated based on a subsequent number of first tags located within the first respective coverage area
410
Iteratively adjust, by the beam adjustment algorithm, the orientation of the first signal beam to a subsequent direction until the first tag threshold is satisfied

**FIG. 4**

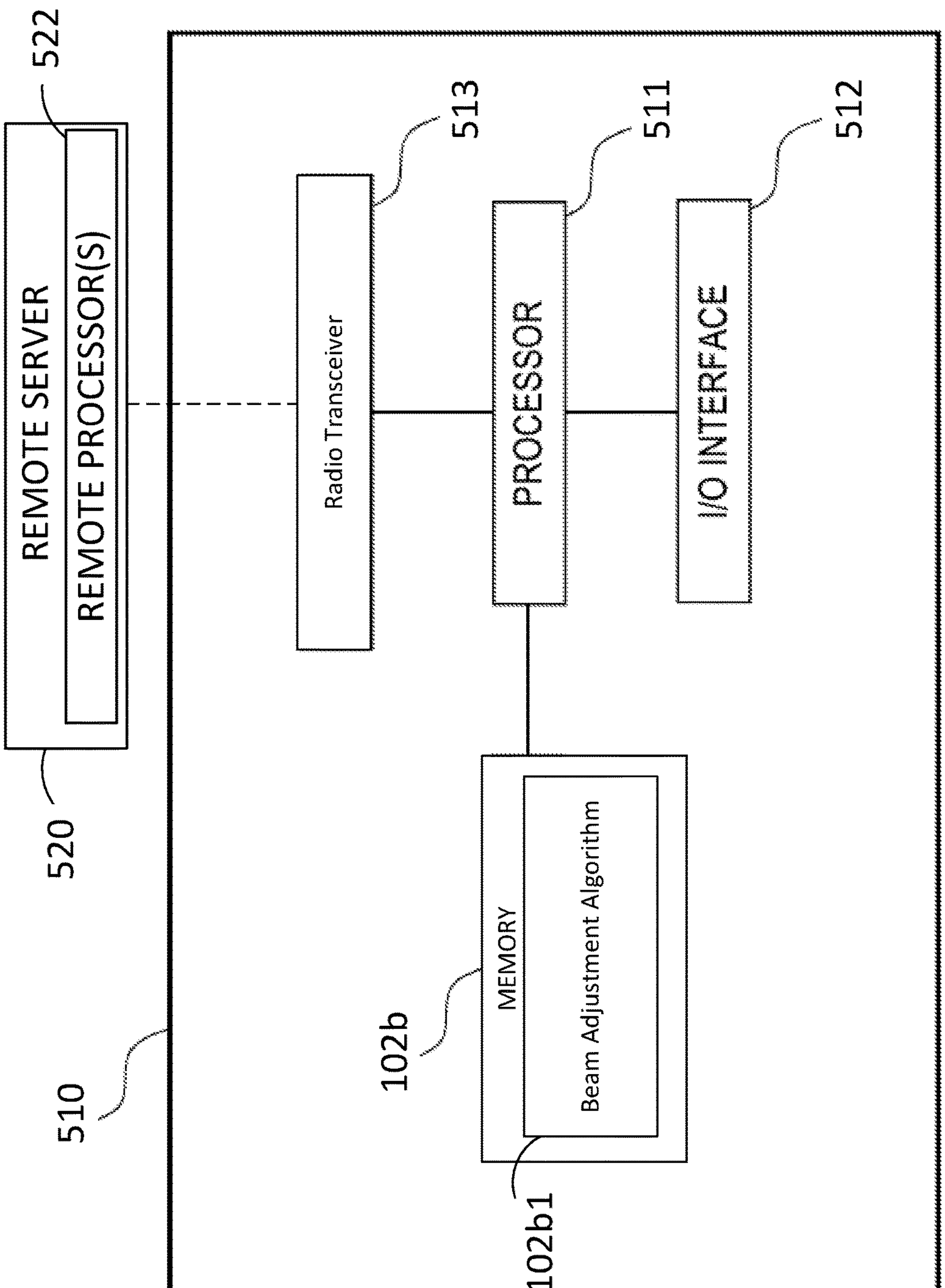
522
REMOTE SERVER
REMOTE PROCESSOR(S)
520
513
Radio Transceiver
511
PROCESSOR
512
I/O INTERFACE
510
102b
MEMORY
Beam Adjustment Algorithm
102b1

FIG. 5

# TECHNIQUES FOR AUTOMATED BEAM ADJUSTMENT FOR PORTAL DIRECTIONALITY

## BACKGROUND

**[0001]** Digital technology is utilized in industrial networks for locationing services. More specifically, location data insight is a relevant component of entities seeking to track locations and statuses of their assets, enhance the productivity of their workers, and to generally optimize workflows. As such, development of systems and devices that reliably provide cost-effective, proximity-based asset visibility solutions are a topic of great interest in the field of industrial networking.

**[0002]** However, conventional locationing systems suffer from several drawbacks that prevent them from providing such reliable and effective locationing services. Namely, many conventional locationing systems rely on hardware that is sub-optimally configured and can therefore produce inaccurate results and/or otherwise fail to focus on the relevant assets. For example, a conventional locationing system may utilize static signal beam orientations from a reader to track the locations of dynamic/moving tags. These static beam orientations may be insufficient to maximize readings of such dynamic/moving tags because static ("stray") tags may be located within the field of view (also referenced herein as a "coverage area") of the static beams. Consequently, conventional locationing systems suffer from issues that minimize the accuracy and effectiveness of dynamic tag tracking.

**[0003]** Thus, there is a need for techniques for automated beam adjustment for portal directionality that allows for accurate dynamic tag tracking by a locationing system.

## SUMMARY

**[0004]** In some aspects, the techniques described herein relate to an assembly including: a transceiver configured to emit a plurality of signal beams that each have a respective coverage area and are oriented in a respective direction; one or more processors; and one or more memories communicatively coupled to the one or more processors storing a beam adjustment algorithm and instructions that, when executed by the one or more processors, cause the assembly to: transmit, by the transceiver, a first signal beam of the plurality of signal beams to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction, determine, by the beam adjustment algorithm, that a first tag threshold is violated based on a number of first tags located within the first respective coverage area, and adjust, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction.

**[0005]** In some aspects, the techniques described herein relate to an assembly, wherein a second number of first tags located within the first respective coverage area of the first signal beam while oriented in the second direction is less than the number of first tags located within the first respective coverage area of the first signal beam while oriented in the first direction.

**[0006]** In some aspects, the techniques described herein relate to an assembly, wherein the first direction and the second direction include a respective azimuth component and a respective elevation component, and the instructions, when executed by the one or more processors, further cause the assembly to adjust the orientation of the first signal beam by: adjusting a first respective azimuth component of the first direction to a second respective azimuth component of the second direction; or adjusting a first respective elevation component of the first direction to a second respective elevation component of the second direction.

**[0007]** In some aspects, the techniques described herein relate to an assembly, wherein the first tags correspond to stray tags, a second number of dynamic tags are also located within the first respective coverage area while the first signal beam is oriented in the first direction, and the first tag threshold corresponds to a ratio of stray tags to dynamic tags.

**[0008]** In some aspects, the techniques described herein relate to an assembly, wherein after adjusting the orientation of the first signal beam to the second direction, the instructions, when executed by the one or more processors, further cause the assembly to: (a) determine, by the beam adjustment algorithm, that the first tag threshold is violated based on a subsequent number of first tags located within the first respective coverage area; (b) adjust, by the beam adjustment algorithm, the orientation of the first signal beam to a subsequent direction; and iteratively perform steps (a) and (b) until the first tag threshold is satisfied.

**[0009]** In some aspects, the techniques described herein relate to an assembly, wherein the instructions, when executed by the one or more processors, further cause the assembly to: determine, by the beam adjustment algorithm, that (i) the second direction or (ii) the first respective coverage area of the first signal beam fails to satisfy a coverage threshold relative to a second signal beam of the plurality of signal beams; and deactivate the first signal beam.

**[0010]** In some aspects, the techniques described herein relate to an assembly, wherein first tags correspond to stray tags, and the instructions, when executed by the one or more processors, further cause the assembly to: receive, by the transceiver, a response signal from a respective tag at a first time indicating that the respective tag is located within the first respective coverage area; receive, by the transceiver, a subsequent response signal from the respective tag at a second time that is different from the first time indicating that the respective tag is located within the first respective coverage area; and determine, by the beam adjustment algorithm, that the respective tag is a stray tag.

**[0011]** In some aspects, the techniques described herein relate to an assembly, wherein the instructions, when executed by the one or more processors, further cause the assembly to: receive, by the transceiver, a second response signal from a second respective tag at the first time indicating that the second respective tag is located within the first respective coverage area; receive, by the transceiver, a third response signal from the second respective tag at the second time indicating that the respective tag is located within a second respective coverage area of a second signal beam of the plurality of signal beams; and determine, by the beam adjustment algorithm, that the second respective tag is an dynamic tag.

**[0012]** In some aspects, the techniques described herein relate to a method including: transmitting, by a transceiver configured to emit a plurality of signal beams that each have a respective coverage area and are oriented in a respective direction, a first signal beam of the plurality of signal beams to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction; determining, by a beam adjustment algorithm, that a first tag threshold is violated based on a number of first tags located within the first respective coverage area; and adjusting, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction.

**[0013]** In some aspects, the techniques described herein relate to a method, wherein a second number of first tags located within the first respective coverage area of the first signal beam while oriented in the second direction is less than the number of first tags located within the first respective coverage area of the first signal beam while oriented in the first direction.

**[0014]** In some aspects, the techniques described herein relate to a method, wherein the first direction and the second direction include a respective azimuth component and a respective elevation component, and the method further includes adjusting the orientation of the first signal beam by: adjusting a first respective azimuth component of the first direction to a second respective azimuth component of the second direction; or adjusting a first respective elevation component of the first direction to a second respective elevation component of the second direction.

**[0015]** In some aspects, the techniques described herein relate to a method, wherein the first tags correspond to stray tags, a second number of dynamic tags are also located within the first respective coverage area while the first signal beam is oriented in the first direction, and the first tag threshold corresponds to a ratio of stray tags to dynamic tags.

**[0016]** In some aspects, the techniques described herein relate to a method, wherein after adjusting the orientation of the first signal beam to the second direction, the method further includes: (a) determining, by the beam adjustment algorithm, that the first tag threshold is violated based on a subsequent number of first tags located within the first respective coverage area; (b) adjusting, by the beam adjustment algorithm, the orientation of the first signal beam to a subsequent direction; and iteratively performing steps (a) and (b) until the first tag threshold is satisfied.

**[0017]** In some aspects, the techniques described herein relate to a method, further including: determining, by the beam adjustment algorithm, that (i) the second direction or (ii) the first respective coverage area of the first signal beam fails to satisfy a coverage threshold relative to a second signal beam of the plurality of signal beams; and deactivating the first signal beam.

**[0018]** In some aspects, the techniques described herein relate to a method, wherein first tags correspond to stray tags, and the method further includes: receiving, by the transceiver, a response signal from a respective tag at a first time indicating that the respective tag is located within the first respective coverage area; receiving, by the transceiver, a subsequent response signal from the respective tag at a second time that is different from the first time indicating that the respective tag is located within the first respective coverage area; and determining, by the beam adjustment algorithm, that the respective tag is a stray tag.

**[0019]** In some aspects, the techniques described herein relate to a method, further including: receiving, by the transceiver, a second response signal from a second respective tag at the first time indicating that the second respective tag is located within the first respective coverage area; receiving, by the transceiver, a third response signal from the second respective tag at the second time indicating that the respective tag is located within a second respective coverage area of a second signal beam of the plurality of signal beams; and determining, by the beam adjustment algorithm, that the second respective tag is an dynamic tag.

**[0020]** In some aspects, the techniques described herein relate to a tangible machine-readable medium including instructions that, when executed, cause a machine to at least: transmit, by a transceiver configured to emit a plurality of signal beams that each have a respective coverage area and are oriented in a respective direction, a first signal beam of the plurality of signal beams to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction; determine, by a beam adjustment algorithm, that a first tag threshold is violated based on a number of first tags located within the first respective coverage area; and adjust, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction.

**[0021]** In some aspects, the techniques described herein relate to a tangible machine-readable medium, wherein the first direction and the second direction include a respective azimuth component and a respective elevation component, and the instructions, when executed, further cause the machine to adjust the orientation of the first signal beam by: adjusting a first respective azimuth component of the first direction to a second respective azimuth component of the second direction; or adjusting a first respective elevation component of the first direction to a second respective elevation component of the second direction.

**[0022]** In some aspects, the techniques described herein relate to a tangible machine-readable medium, wherein the first tags correspond to stray tags, a second number of dynamic tags are also located within the first respective coverage area while the first signal beam is oriented in the first direction, and the first tag threshold corresponds to a ratio of stray tags to dynamic tags.

**[0023]** In some aspects, the techniques described herein relate to a tangible machine-readable medium, wherein after adjusting the orientation of the first signal beam to the second direction, the instructions, when executed, further cause the machine to: (a) determine, by the beam adjustment algorithm, that the first tag threshold is violated based on a subsequent number of first tags located within the first respective coverage area; (b) adjust, by the beam adjustment algorithm, the orientation of the first signal beam to a subsequent direction; and iteratively perform steps (a) and (b) until the first tag threshold is satisfied.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0024]** The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention, and explain various principles and advantages of those embodiments.

**[0025]** FIG. **1** depicts an example environment in which systems/devices for automated beam adjustment for portal directionality may be implemented, in accordance with various embodiments described herein.

**[0026]** FIG. **2**A depicts an example signal beam configuration for automated beam adjustment for portal directionality, in accordance with various embodiments described herein.

**[0027]** FIG. **2**B depicts an example signal beam adjustment relative to the configuration of FIG. **2**A, in accordance with various embodiments described herein.

**[0028]** FIG. **3**A depicts a first example asset tracking scenario where a signal beam is adjusted to reduce the number of stray tags within the signal beam coverage area, in accordance with various embodiments described herein.

**[0029]** FIG. **3**B depicts a second example asset tracking scenario where one or more signal beams are adjusted to reduce the number of stray tags within the signal beam coverage area, in accordance with various embodiments described herein.

**[0030]** FIG. **4** is a flowchart representative of a method for automated beam adjustment for portal directionality, in accordance with embodiments described herein.

**[0031]** FIG. **5** is a block diagram of an example logic circuit for implementing example methods and/or operations described herein.

**[0032]** Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

**[0033]** The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

## DETAILED DESCRIPTION

**[0034]** As previously mentioned, conventional locationing systems generally suffer from accuracy and efficiency issues stemming from, inter alia, insufficiently configured and/or configurable hardware and a corresponding lack of stray tag suppression. For example, many conventional locationing systems utilize static beam signal configurations/orientations and/or unsteerable antennas to track dynamic/moving tags. However, these static beam signal configurations/orientations, and the resulting tracked tags, often include a significant number of stray tags that can reduce the likelihood that the tags of interest (i.e., dynamic tags) are identified and tracked.

**[0035]** To illustrate, a conventional reader may output a static signal beam with a coverage area that is oriented in a direction to partially include a set of static assets (e.g., shelves with stationary products at a grocery store, wall of stacked boxes in an unloaded trailer) and partially include a pathway where assets may be transported. This static signal beam may occasionally poll tags within the coverage area associated with assets being transported along the pathway but will nearly always poll tags associated with the static assets, as they are always within the coverage area. Ideally, the conventional reader would not poll and/or otherwise monitor these static tag locations and instead prioritize processing resources to accurately track locations of the dynamic tags being transported along the pathway. Nevertheless, conventional locationing systems frequently feature static signal beam configurations that either overrepresent static tags and/or are otherwise not positioned along pathways to capture dynamic tags of interest. Accordingly, in general, conventional locationing systems can struggle to accurately track dynamic tags.

**[0036]** It is an objective of the present disclosure to eliminate these and other problems with conventional locationing systems via techniques for automated beam adjustment for portal directionality that may accurately track dynamic tags through a beam adjustment algorithm. In particular, the techniques of the present disclosure alleviate the issues present with conventional systems/devices by essentially determining that stray tags are overrepresented in the polling results for a given signal beam and thereafter adjusting the orientation of that given signal beam to reduce the erroneous capture of such stray tags. As a result, the techniques of the present disclosure accurately track dynamic tags (and their associated assets) with an accuracy that was previously unachievable using conventional techniques.

**[0037]** Thus, in accordance with the above, and with the disclosure herein, the present disclosure includes improvements in computer functionality or in improvements to other technologies at least because the present disclosure describes that, e.g., locationing systems, and their related various components, may be improved or enhanced with the disclosed beam adjustment algorithm that provides more accurate locationing/tracking services for dynamic tags and corresponding assets. That is, the present disclosure describes improvements in the functioning of a locationing system itself or "any other technology or technical field" (e.g., the field of distributed/industrial locationing systems) because the disclosed beam adjustment algorithm improves and enhances operation of locationing systems by introducing dynamic signal beam orientation/direction adjustments to eliminate/reduce erroneous static tag logging and other inefficiencies typically experienced over time by locationing systems lacking such a beam adjustment algorithm. This improves the state of the art at least because such previous locationing systems are inaccurate as they lack the ability for dynamically adjusting signal beam orientations/directions in the manners described herein.

**[0038]** In addition, the present disclosure includes applying various features and functionality, as described herein, with, or by use of, a particular machine, e.g., a tag, a reader, a server, and/or other hardware components as described herein.

**[0039]** Moreover, the present disclosure includes specific features other than what is well-understood, routine, conventional activity in the field, or adding unconventional steps that demonstrate, in various embodiments, particular useful applications, e.g., transmitting, by a transceiver, a first signal beam of the plurality of signal beams to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction; determining, by a beam adjustment algorithm, that a first tag threshold is violated based on a number of first tags located within the first respective coverage area, and/or adjusting, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction, among others.

**[0040]** Turning to the figures, FIG. **1** depicts an example environment **100** in which systems/devices for automated beam adjustment may be implemented, in accordance with embodiments described herein. The example environment **100** may comprise, include, and/or otherwise be a part of a networking environment in which the systems/devices of the present disclosure may operate. In the example embodiment of FIG. **1**, the example environment **100** includes a reader **102** that is communicatively coupled to a first tag **106***a* of a first asset **106**, a second tag **107***a* of a second asset **107**, a third tag **108***a* of an N$^{th}$ asset **108**, and a server **110**. Generally speaking, the reader **102**, the first tag **106***a*, the second tag **107***a*, the third tag **108***a*, and/or the server **110** may be capable of executing instructions to, for example, implement operations of the example methods described herein, as may be represented by the flowcharts of the drawings that accompany this description. Namely, the reader **102** may be connected to the first tag **106***a*, the second tag **107***a*, the third tag **108***a*, and/or the server **110** across multiple communication channels and may generally be configured to receive and process information received from the first tag **106***a*, the second tag **107***a*, the third tag **108***a*, and/or the server **110**.

**[0041]** The example environment **100** may be or include any suitable real-world environment, such as a grocery store, loading warehouse, hospital, etc., and the area(s) of interest covered by the reader **102** may be or include high travel density asset pathways corresponding to the real-world environment. For example, an area of interest covered by the signal beams of the reader **102** may include an entry/exit pathway to/from a grocery store, where the reader **102** may track dynamic assets as entities enter/exit the store. As another example, an area of interest may be individual loading docks, storage areas, movement pathways for equipment/machinery, etc. within a warehouse.

**[0042]** In any event, the reader **102** stores a tag database **102***b***2** and a locationing engine **102***b***3**, and the server **110** may optionally store a tag database **110***b***1** and a locationing engine **110***b***2**. The tag database **102***b***2**, **110***b***1** may be or include a listing of tags (e.g., tag **106***a*, tag **107***a*, tag **108***a*) that are proximate to specific anchors (e.g., reader **102**) and/or otherwise transmit data to/from the particular anchor (s). More specifically, the tag database **102***b***2**, **110***b***1** listings may include identification information about each of the tags **106***a*, **107***a*, **108***a* and/or the assets **106**, **107**, **108** associated with the tags **106***a*, **107***a*, **108***a*, as well as location information determined by the reader **102**. The tag database **102***b***2**, **110***b***1** may include any suitable information related to the tags and/or the assets associated with the tags.

**[0043]** To update the tag database **102***b***2**, **110***b***1**, the reader **102** may periodically request and/or otherwise receive updates from various tags (e.g., tag **106***a*, tag **107***a*, tag **108***a*) disposed around an environment (e.g., example environment **100**), and the reader **102** may determine (via the one or more processors **102***c*) one or more tags indicated in the received data. The reader **102** may then update the tag listing for each tag **106***a*, **107***a*, **108***a* by inputting the data received from the respective tags **106***a*, **107***a*, **108***a* into the corresponding tag listing of the tag database **102***b***2**, **110***b***1**. For example, the tag database **102***b***2**, **110***b***1** may indicate at a first time that the reader **102** received data from the first tag **106***a* and the second tag **107***a*. At a second time, the reader **102** may transmit a request to and/or may otherwise receive an update from proximate tags indicating that the reader **102** received/captured data from the first tag **106***a*, the second tag **107***a*, and the third tag **108***a*. Thus, the entries of the tag database **102***b***2**, **110***b***1** may indicate that the N$^{th}$ asset **108** moved into a receptive proximity of the reader **102** at some point between the first time and the second time, such that the reader **102** was able to receive data transmitted from the third tag **108***a* at the second time.

**[0044]** In certain embodiments, the first asset **106**, the second asset **107**, and/or the N$^{th}$ asset **108** may be static assets or dynamic assets. Continuing the prior example, the first asset **106** and the second asset **107** may be static assets within the coverage area of one or more signal beams of the reader **102**, as indicated by the received data from both tags **106***a*, **107***a* at the first time and the second time. The N$^{th}$ asset **108** may be a dynamic asset within the coverage area of the one or more signal beams of the reader **102**, as indicated by the reader **102** receiving data from the N$^{th}$ asset tag **108***a* only at the second time. Accordingly, in this example, the first tag **106***a* and the second tag **107***a* may be static tags and the N$^{th}$ tag **108***a* may be a dynamic tag with respect to the reader **102**.

**[0045]** As referenced herein, a "coverage area" of a signal beam may generally refer to the cross-sectional area or volume covered by the signal beam, as emitted by the radio transceiver **102***a*. These signal beams may generally have circular cross-sectional profiles of any suitable dimensions, but it should be appreciated that the signal beams may be output through guides and/or other suitable devices to shape the signal beams into any desired cross-sectional profile. For example, the reader **102** may be disposed on a ceiling near an entrance/exit (i.e., a "portal") of a grocery store, and a first signal beam emitted by the reader **102** may have a coverage area with a one-meter diameter at the floor of the portal. Thus, the total coverage area of the first signal beam may include a three-dimensional cone extending from a point at the radio transceiver **102***a* to the one-meter diameter circle at the portal floor.

**[0046]** Regardless, the reader **102** may also include a locationing engine **102***b***3**, and the server **110** may optionally include a locationing engine **110***b***2**, which generally analyzes data received from the tags **106***a*, **107***a*, **108***a* to update locations of the tags **106***a*, **107***a*, **108***a*. More specifically, the locationing engine **102***b***3**, **110***b***2** may analyze data packets and/or information extracted and/or otherwise derived from the data packets to determine a direction of travel along which each tag **106***a*, **107***a*, **108***a* is currently traveling. Such data packets may generally be or include identification information corresponding to the tags **106***a*, **107***a*, **108***a* and/or the associated assets **106**, **107**, **108**, remaining battery life information, asset-specific data payloads, and/or any other suitable data or combinations thereof. The reader **102** may transmit polling requests or query repeat commands to receive data from the tags **106***a*, **107***a*, **108***a* at any suitable transmission rate such as every several milliseconds, every second, several seconds, minutes, and/or any other suitable frequency. In certain embodiments, data packets transmitted to/from any particular tag **106***a*, **107***a*, **108***a* may be or include an ultra-high frequency (UHF) radio-frequency identification (RFID) signal and/or any other suitable data packets or signals.

**[0047]** When the locationing engine **110***b***2** determines the current location/position for a particular tag, the locationing engine **110***b***2** may also analyze prior data transmitted from the particular tag to determine whether the current location/position is different from a prior location/position determined from the prior data. For example, the locationing engine **110***b***2** may receive data from a first tag and determine that the first tag is estimated to be in a first location, as registered by a first signal beam from the reader **102**. The locationing engine **110*b*2** may then analyze prior data corresponding to the first tag (e.g., as stored in the tag database **110*b*1**) and determine that the last report from the first tag indicated that the first tag was located in a second (e.g., prior) location/position at the time the last report was transmitted from the first tag, as registered by a second signal beam from the reader **102**. Consequently, the locationing engine **110*b*2** may determine that the first tag has changed location from the second location/position to the first location/position during the time interval that elapsed between the last report and the current report from the first tag. However, it should be appreciated that the locationing engine **110*b*2** may determine differences in location/position for an individual tag based on multiple signals received from a single signal beam over time. Additionally, it should be appreciated that the beam adjustment algorithm **102*b*1** may perform some/all of the functions described herein as performed by the locationing engine **110*b*2**, and vice versa.

**[0048]** As another example, the locationing engine **110*b*2** may receive data from a second tag and determine that the second tag is estimated to be in a third location, as registered by a third signal beam from the reader **102**. The locationing engine **110*b*2** may then analyze prior data corresponding to the second tag (e.g., as stored in the tag database **110*b*1**) and determine that the last report from the second tag indicated that the second tag was located in the third location at the time the last report was transmitted from the second tag. Consequently, the locationing engine **110*b*2** may determine that the second tag has remained in the third location during the time interval that elapsed between the last report and the current report from the second tag, and may identify the second tag as a static tag. The locationing engine **110*b*2** may transmit this information to the beam adjustment algorithm **102*b*1**, which may determine that the orientation/direction of the third signal beam should be adjusted to remove the second tag from the coverage area of the third signal beam.

**[0049]** As part of these communications between the server **110** and the individual tags **106*a***, **107*a***, **108*a***, the reader **102** may receive data from the server **110** and capture data from the tags **106*a***, **107*a***, **108*a***, and may transmit communications based on that received/captured data to the server **110** and/or the tags **106*a***, **107*a***, **108*a***. Broadly speaking, the reader **102** may be configured to transmit and receive data to/from the server **110** and nearby tags (e.g., the first tag **106*a***, the second tag **107*a***, the third tag **108*a***). In certain embodiments, the reader **102** may be an UHF RFID reader device that communicates with some/all of the devices in the environment **100** via UHF radio signals. In some embodiments, the reader **102** may be a device that executes and/or conforms to any suitable software operating system (e.g., Android, IOS), a custom Internet of Things (IoT) bridge device with a UHF radio, and/or any other suitable device or combination thereof.

**[0050]** Namely, the reader **102** may be configured to periodically listen for data packets from nearby tags (e.g., tags **106*a***, **107*a***, **108*a***), transmit the data packets and/or data obtained therein to the server **110**, and/or broadcast requests received from the server **110** to such nearby tags. As an example, the reader **102** may receive requests from the server **110**, and may subsequently transmit requests to proximate tags **106*a***, **107*a***, **108*a*** based on the requests. Such requests from the server **110** may be or include instructions causing the tags **106*a***, **107*a***, **108*a*** to transmit identification data to the reader **102** and/or other suitable instructions or combinations thereof.

**[0051]** The reader **102** may also transmit and receive data (e.g., data packets) to/from any of the tags **106*a***, **107*a***, **108*a*** and to calculate signal beam direction/orientation adjustments based on the data received from those tags **106*a***, **107*a***, **108*a*** by executing the beam adjustment algorithm **102*b*1** stored in memory **102*b***. The reader **102** may then transmit these signal beam orientation/direction adjustments, along with some/all of the received data from the tags **106*a***, **107*a***, **108*a***, to the server **110** for tracking and/or otherwise notifying a user (e.g., a locationing system administrator) of a location/position of an asset **106**, **107**, **108** associated with the tags **106*a***, **107*a***, **108*a***.

**[0052]** As referenced herein, the signal beam orientation/direction adjustments determined and implemented as a result of the beam adjustment algorithm **102*b*1** may be or include any suitable adjustments to the orientation, directionality, and/or configuration of an individual signal beam and/or a set of signal beams emitted by the reader **102**. For example, a first adjustment determined by the beam adjustment algorithm **102*b*1** may cause the reader **102** to reposition the coverage area of a first signal beam by five degrees in any suitable direction (e.g., positive/negative azimuth, positive/negative elevation). As a second example, a second adjustment determined by the beam adjustment algorithm **102*b*1** may cause the reader **102** to reposition the coverage area of a second signal beam and a third signal beam by ten degrees in similar/identical directions or dissimilar/opposite directions.

**[0053]** As a third example, a third adjustment determined by the beam adjustment algorithm **102*b*1** may cause the reader **102** to deactivate a fourth signal beam because repositioning the coverage area of the fourth signal beam in an optimal direction (e.g., positive azimuth) by an optimal degree (e.g., six degrees) would result in a coverage area overlap between the fourth signal beam and a fifth signal beam that violates, exceeds, and/or otherwise fails to satisfy a coverage overlap threshold. This deactivation may therefore minimize coverage area redundancy of emitted signal beams.

**[0054]** Moreover, as illustrated in the prior examples, it should be appreciated that the "optimal" direction and/or degree determined and implemented by the beam adjustment algorithm **102*b*1** to adjust signal beam orientation/direction may be any suitable direction and/or degree or combinations thereof. Namely, an optimal direction adjustment determined by the beam adjustment algorithm **102*b*1** may include an adjustment in an isolated direction (e.g., only azimuth, only elevation) or a composite direction and may include an adjustment of any suitable degree value. For example, a first adjustment determined by the beam adjustment algorithm **102*b*1** for a first signal beam may cause the radio transceiver **102*a*** to adjust the coverage area of the first signal beam by two degrees in the positive azimuth direction and three degrees in the negative elevation direction.

**[0055]** Generally, the beam adjustment algorithm **102*b*1** may be or include instructions causing the processors **102*c*** of the reader **102** to determine that a stray tag threshold has been violated for a signal beam and corresponding adjustment(s) to the direction/orientation of the signal beam. The stray tag threshold may be or include a percentage or ratio of the stray tags to dynamic tags or to total tags within the signal beam coverage area. In other words, if the stray tag threshold is violated, then the associated signal beam coverage area is currently located over a larger number of stray tags than is acceptable to achieve high accuracy/granularity dynamic tag tracking. Accordingly, the beam adjustment algorithm **102*b*1** may determine and cause the radio transceiver **102*a*** to implement an adjustment to the signal beam orientation/direction to reduce the number of stray tags included within the coverage area.

**[0056]** As part of determining/evaluating the stray tag threshold, the beam adjustment algorithm **102*b*1** may first analyze tag response signals for proximate tags over time to determine which tags within a particular signal beam coverage area is/are stray tags. The beam adjustment algorithm **102*b*1** may analyze consecutive responses from tags within signal beam coverage areas to determine whether the period during which an individual tag has remained within a signal beam coverage area exceeds a period of time threshold. If the algorithm **102*b*1** determines that the threshold is exceeded, the algorithm **102*b*1** may identify the corresponding tag as a stray tag.

**[0057]** For example, at a first time, the reader **102** may receive a response signal from a first tag, indicating that the first tag is within the coverage area of a first signal beam emitted from the radio transceiver **102*a***. At a second time, the reader **102** may receive a subsequent response signal from the first tag indicating that the first tag is still located within the coverage area of the first signal beam. The beam adjustment algorithm **102*b*1** may analyze these response signals from the first tag and determine that the first tag has remained within the coverage area of the first signal beam for longer than a threshold period, such that the first tag is likely a stray tag that is statically positioned within the coverage area of the first signal beam. Accordingly, the beam adjustment algorithm **102*b*1** may identify the first tag as a stray tag and may proceed to evaluate whether the stray tag threshold is violated for the first signal beam by including the first tag as one of the number of stray tags.

**[0058]** Once the beam adjustment algorithm **102*b*1** determines that the stray tag threshold is violated for a signal beam, the algorithm **102*b*1** may determine an adjustment to the signal beam orientation/direction to reduce the number of stray tags contained within the coverage area. In certain embodiments, the adjustment may be a pre-determined value, such as five degrees in any suitable direction (e.g., azimuth, elevation). However, the beam adjustment algorithm **102*b*1** may also dynamically generate orientation/direction adjustments based on the estimated locations of the stray tags. The locationing engine **110*b*2** may return estimated locations for the stray tags, and the beam adjustment algorithm **102*b*1** may determine an optimal orientation/direction adjustment for the signal beam that may eliminate most/all stray tags within the signal beam coverage area.

**[0059]** For example, a first signal beam may be oriented such that three stray tags are located within the coverage area at a far edge. The beam adjustment algorithm **102*b*1** may determine that these three stray tags violate the stray tag threshold and determine that the first signal beam orientation/direction should be adjusted. The algorithm **102*b*1** may receive estimated locations of the three stray tags from the locationing engine **110*b*2** and determine that an adjustment of two degrees in the negative azimuthal direction will likely be sufficient to remove the three stray tags from the coverage area. The algorithm **102*b*1** may then generate an instruction for a two-degree negative azimuthal adjustment to the first signal beam, which may cause the radio transceiver **102*a*** to adjust the direction/orientation of the first signal beam by two-degrees in the negative azimuthal direction.

**[0060]** Of course, it should be appreciated that the beam adjustment algorithm **102*b*1** may be or include any suitable instructions that enable the reader **102** and/or any other suitable device(s) (e.g., server **110**) to evaluate stray tag thresholds for signal beams, determine direction/orientation adjustments for the signal beams, and/or perform other suitable adjustments to the signal beam configuration(s) (e.g., turning signal beams off, changing the signal beam configuration).

**[0061]** The reader **102** may also include a radio transceiver **102*a*** configured to transmit/receive data streams to/from various devices of the example environment **100**, such as the server **110** and the tags **106*a***, **107*a***, **108*a***. The radio transceiver **102*a*** may include an antenna with an associated gain profile corresponding to the transceiver's **102*a*** antenna converting input power into radio waves (e.g., transmission) and/or received radio waves into electrical power (e.g., receiving). More specifically, the radio transceiver **102*a*** may be a phased-array antenna configured to steer transmitted and received signal beams in various directions. This phased-array configuration enables the reader **102** and/or the server **110** to accurately determine the bearing (i.e., location in azimuth and elevation) of any tag within the coverage area of signal beams transmitted by the radio transceiver **102*a*** each time the tag is captured by a signal beam.

**[0062]** The assets **106**, **107**, **108** may generally be any device, component, or object that an entity may desire to track and/or otherwise locate. For example, the assets **106**, **107**, **108** may be large and calibrated tools used in and/or for oil and gas equipment/operations, parcels for delivery by a shipping company, hospital equipment that is and/or may be moved to different floors/rooms, wristbands attached to hospital patients, and/or any other suitable objects or combinations thereof. While illustrated as three assets **106**, **107**, **108**, it should be appreciated that the reader **102** may simultaneously communicate with any suitable number of assets **106**, **107**, **108** via the associated tags **106*a***, **107*a***, **108*a***. Thus, the $N^{th}$ asset **108** may be a third asset, a fifth asset, a twentieth asset, a one-hundredth asset, and/or any other integer value asset.

**[0063]** Each asset **106**, **107**, **108** may also include a corresponding tag **106*a***, **107*a***, **108*a*** that may be configured to respond to polling requests by transmitting information associated with the asset via the radio transceiver **106*a*1**, **107*a*1**, **108*a*1** to, for example, the reader **102**. Each asset tag **106*a***, **107*a***, **108*a*** may also include one or more processors **106*a*2**, **107*a*2**, **108*a*2** configured to interpret and/or execute such polling requests and/or other instructions contained in signals received from the reader **102**, server **110**, and/or other suitable device(s). For example, the processors **106*a*2**, **107*a*2**, **108*a*2** may be configured to interpret polling requests and/or other signals received from the reader **102** and thereby transmit data packets to the reader **102**.

**[0064]** Moreover, in certain embodiments, a workstation (not shown) may be communicatively connected to the server **110**, and a user/operator may access the server **110** to retrieve a location associated with an asset **106**, **107**, **108**. The workstation may query the server **110** with the identification tag of the corresponding asset **106**, **107**, **108**, and the server **110** may match the identification tag with a location entry in the tag database **110*b*1** associated with the corresponding asset **106**, **107**, **108**. The server **110** may then forward the location entry to the workstation for viewing by the user/operator.

**[0065]** More generally, the one or more memories **102*b***, **110*b*** may include one or more forms of volatile and/or non-volatile, fixed and/or removable memory, such as read-only memory (ROM), electronic programmable read-only memory (EPROM), random access memory (RAM), erasable electronic programmable read-only memory (EEPROM), and/or other hard drives, flash memory, MicroSD cards, and others. In general, a computer program or computer based product, application, or code (e.g., beam adjustment algorithm **102*b*1**, and/or other computing instructions described herein) may be stored on a computer usable storage medium, or tangible, non-transitory computer-readable medium (e.g., standard random access memory (RAM), an optical disc, a universal serial bus (USB) drive, or the like) having such computer-readable program code or computer instructions embodied therein, wherein the computer-readable program code or computer instructions may be installed on or otherwise adapted to be executed by the one or more processors **102*c***, **110*a*** (e.g., working in connection with a respective operating system in the one or more memories **102*b***, **110*b***) to facilitate, implement, or perform the machine readable instructions, methods, processes, elements or limitations, as illustrated, depicted, or described for the various flowcharts, illustrations, diagrams, figures, and/or other disclosure herein.

**[0066]** In this regard, the program code may be implemented in any desired program language, and may be implemented as machine code, assembly code, byte code, interpretable source code or the like (e.g., via Golang, Python, C, C++, C#, Objective-C, Java, Scala, ActionScript, JavaScript, HTML, CSS, XML, etc.). Moreover, the one or more memories **102*b***, **110*b*** may also store machine readable instructions, including any of one or more application(s), one or more software component(s), and/or one or more APIs, which may be implemented to facilitate or perform the features, functions, or other disclosure described herein, such as any methods, processes, elements or limitations, as illustrated, depicted, or described for the various flowcharts, illustrations, diagrams, figures, and/or other disclosure herein.

**[0067]** The one or more processors **102*c***, **110*a*** may be connected to the one or more memories **102*b***, **110*b*** via a computer bus (not shown) responsible for transmitting electronic data, data packets, or otherwise electronic signals to and from the one or more processors **102*c***, **110*a*** and one or more memories **102*b***, **110*b*** to implement or perform the machine readable instructions, methods, processes, elements or limitations, as illustrated, depicted, or described for the various flowcharts, illustrations, diagrams, figures, and/or other disclosure herein.

**[0068]** The one or more processors **102*c***, **110*a*** may interface with the one or more memories **102*b***, **110*b*** via the computer bus to execute any suitable application or executable instructions (e.g., beam adjustment algorithm **102*b*1**) necessary to perform any of the actions associated with the methods of the present disclosure. The one or more processors **102*c***, **110*a*** may also interface with the one or more memories **102*b***, **110*b*** via the computer bus to create, read, update, delete, or otherwise access or interact with the data stored in the one or more memories **102*b***, **110*b*** and/or external databases (e.g., a relational database, such as Oracle, DB2, MySQL, or a NoSQL based database, such as MongoDB). The data stored in the one or more memories **102*b***, **110*b*** and/or an external database may include all or part of any of the data or information described herein, including, for example, asset tag **106*a***, **107*a***, **108*a*** data packets, asset location data, beam adjustment data/thresholds, stray tag thresholds, and/or other suitable information or combinations thereof.

**[0069]** The radio transceivers **102*a***, **106*a*1**, **107*a*1**, **108*a*1** and the networking interface **110*c*** may be configured to communicate (e.g., send and receive) data via one or more external/network port(s) to one or more networks or local terminals, as described herein. In some embodiments, the radio transceivers **102*a***, **106*a*1**, **107*a*1**, **108*a*1** and/or the networking interface **110*c*** may include a client-server platform technology such as ASP.NET, Java J2EE, Ruby on Rails, Node.js, a web service or online API, responsive for receiving and responding to electronic requests. The radio transceivers **102*a***, **106*a*1**, **107*a*1**, **108*a*1** and/or the networking interface **110*c*** may implement the client-server platform technology that may interact, via the computer bus, with the one or more memories **102*b***, **110*b*** (including the applications(s), component(s), API(s), data, etc. stored therein) to implement or perform the machine readable instructions, methods, processes, elements or limitations, as illustrated, depicted, or described for the various flowcharts, illustrations, diagrams, figures, and/or other disclosure herein.

**[0070]** According to some embodiments, the radio transceivers **102*a***, **106*a*1**, **107*a*1**, **108*a*1** and/or the networking interface **110*c*** may include, or interact with, one or more transceivers (e.g., WWAN, WLAN, and/or WPAN transceivers) functioning in accordance with IEEE standards, 3GPP standards, or other standards, and that may be used in receipt and transmission of data via external/network ports connected to a network. In some embodiments, the network (not shown) may comprise a private network or local area network (LAN). Additionally, or alternatively, the network may comprise a public network such as the Internet. In some embodiments, the network may comprise routers, wireless switches, or other such wireless connection points communicating to the server **110** (via networking interface **110*c***) via wireless communications based on any one or more of various wireless standards, including by non-limiting example, an RFID standard, a BLUETOOTH standard (e.g., BLE), IEEE 802.11a/b/c/g (WIFI), or the like.

**[0071]** To illustrate a scenario where a reader and asset tag(s) may transmit and/or receive data packets and determine signal beam adjustments from those data packets, FIG. **2**A depicts an example signal beam configuration **200** for automated beam adjustment for portal directionality, in accordance with various embodiments described herein. In particular, the example configuration **200** illustrated in FIG. **2**A depicts a reader **202** transmitting a plurality of signal beams **204*a-f***, which for the purposes of simplicity, are illustrated in FIG. **2**A by their cross-sectional profiles at ground level. The example configuration **200** further includes four zones **206*a-d***. The first zone **206*a*** and the second zone **206*b*** may be zones of interest, and the third zone **206*c*** and the fourth zone **206*d*** may not be zones of interest. For example, the first zone **206*a*** and the second zone **206*b*** may represent a pathway leading to/from an entrance/exit portal of a department store, where the first zone **206***a* is outside the store and the second zone **206***b* is inside the store.

**[0072]** Each signal beam **204***a-f* has a current orientation/direction, as illustrated by the parenthetical numbers proximate to each signal beam **204***a-f*. The orientation/direction for a particular signal beam may be in terms of any suitable dimensions, such as degrees of azimuth and elevation. For example, the first signal beam **204***a* has an orientation/direction of zero degrees in the azimuthal direction and 45 degrees in the elevation direction. Similarly, the second signal beam **204***b* has an orientation/direction of zero degrees in the azimuthal direction and 30 degrees in the elevation direction, the third signal beam **204***c* has an orientation/direction of 180 degrees in the azimuthal direction and 30 degrees in the elevation direction, the fourth signal beam **204***d* has an orientation/direction of 180 degrees in the azimuthal direction and 45 degrees in the elevation direction, the fifth signal beam **204***e* has an orientation/direction of 270 degrees in the azimuthal direction and 15 degrees in the elevation direction, and the sixth signal beam **204***f* has an orientation/direction of 90 degrees in the azimuthal direction and 15 degrees in the elevation direction.

**[0073]** As previously mentioned, the signal beams emitted by a reader may have any suitable configuration, such as the specific configuration **200** illustrated in FIG. **2**A. As referenced herein, a "signal beam configuration" may generally refer to the orientation(s)/direction(s) of all signal beams emitted by a reader (e.g., reader **202**) at a particular time. For example, the signal beam configuration **200** emitted by the reader **202** features four signal beams **204***a*, **204***b*, **204***c*, **204***d* emitted along a line extending from a first zone **206***a* into a second zone **206***b* and two signal beams **204***e*, **204***f* straddling the boundary between the first zone **206***a* and the second zone **206***b*. In certain embodiments, the reader **202** may have multiple pre-programmed signal beam configurations to cover the path from the second zone **206***b* (e.g., "inside") to the first zone **206***a* (e.g., "outside"). Regardless, each of the signal beam configurations is intended to cover the zones of interest (e.g., first zone **206***a*, second zone **206***b*) well enough to read dynamic tags at least twice in each zone of interest as the dynamic tags move through the coverage areas of the reader **202** while not dwelling too long in one area, and thereby avoid missing other dynamic tags.

**[0074]** More specifically, the signal beam configuration **200** of FIG. **2**A may be intended to read dynamic tags in the first zone **206***a* (e.g., "outside") and then in the second zone **206***b* (e.g., "inside"), or vice-versa, and report the direction of travel between the zones **206***a*, **206***b*. The accuracy of a directionality estimation performed by the reader **202** and/or other suitable device(s) is generally a function of the number of times a dynamic tag is read as the tag traverses the coverage areas of the reader **202**. In other words, the more frequently the dynamic tag is read as it passes by the reader **202**, the more likely the reader **202** will have enough bearing estimate samples to produce a correct directionality estimate. Conversely, the less frequently the dynamic tag is read as it passes by the reader **202**, the less likely the reader **202** will have enough bearing estimate samples to produce a correct directionality estimate.

**[0075]** Accordingly, optimally positioning the signal beams **204***a-f* within the zones of interest (e.g., first zone **206***a*, second zone **206***b*) is of paramount importance to ensure that dynamic tags are read as frequently as possible. As previously mentioned, reducing the number of stray tag reads greatly advances these efforts to increase dynamic tag read frequency, and an example signal beam adjustment **220** configured to reduce stray tag reads is illustrated in FIG. **2**B. In particular, FIG. **2**B depicts an example signal beam adjustment **220** relative to the configuration **200** of FIG. **2**A, in accordance with various embodiments described herein.

**[0076]** When stray tags are within the coverage areas of the signal beams **204***a-f*, the reader **202** will spend time reading these stray tags and reduce the time the reader **202** can read the tags of interest (e.g., dynamic tags). As a result, the reader **202** will capture fewer signals and/or otherwise data from the dynamic tags as they travel through the coverage areas of the signal beams **204***a-f* and generate less accurate travel direction determinations of the dynamic tags. Thus, the challenge addressed by the example signal beam adjustment **220** is to limit the stray tag readings while still allowing the dynamic tags to be read enough times to allow the reader **202** and/or other suitable device(s) to make accurate travel direction determinations for the dynamic tags.

**[0077]** Typically, stray tags are not in the path that dynamic tags will take, and the stray tags are on the periphery of the coverage areas of the signal beam configuration. Therefore, the operating principle of the example signal beam adjustment **220** is to adjust any of the signal beams **204***a-f* that capture data from too many stray tags towards the presumed dynamic tag path in a manner that reduces the number of stray tags being read and gives more opportunity for reading the dynamic tags.

**[0078]** For example, assume that the first signal beam **204***a* and the fourth signal beam **204***d* have both captured data from too many stray tags (i.e., violated the stray tag threshold), such that the reader **202** and/or other suitable device(s) determines that the orientation/direction of the signal beams **204***a*, **204***d* should be adjusted. In this example, the orientations/directions of the first signal beam **204***a* and the fourth signal beam **204***d* may be adjusted (as represented by the arrows **222***a*, **222***b*) by five degrees each. The new orientation/direction of the first signal beam **204***a* may be zero degrees in the azimuthal direction and 40 degrees in the elevation direction, and the new orientation/direction of the fourth signal beam **204***d* may be 180 degrees in the azimuthal direction and 40 degrees in the elevation direction. Of course, it should be appreciated that the orientation/direction of the signal beams may be adjusted by any suitable amount.

**[0079]** Even such small orientation/direction adjustments to signal beams can yield significant improvements to the ratio of stray-to-dynamic tag reads. To illustrate, real-world tests similar to the prior example featuring adjustments to the orientation/direction of signal beams (e.g., signal beams **204***a*, **204***d*) by five degrees resulted in a reduction of the number of total unique tags read by a reader from **43** to **22**. This nearly 50% decrease to the number of unique tag reads represented a substantial decrease in the number of stray tags read by the reader, which correspondingly represented a significant increase in the available time for the reader to capture/read dynamic tags.

**[0080]** Moreover, while described herein primarily as a singular process, the orientation/direction adjustments to signal beams may be performed iteratively until the reader **202** and/or other device(s) determine that the signal beams are each optimally oriented. For example, the reader **202**

may adjust the orientation/direction of the fourth signal beam **204***d* five degrees, as illustrated in FIG. **2**B, and may subsequently determine that the orientation/direction of the fourth signal beam **204***d* should be further adjusted by another five degrees and/or any other suitable value(s) (e.g., one degree, two degrees, ten degrees, etc.) in any suitable direction(s). The reader **202** may iteratively adjust the orientation/direction of the fourth signal beam **204***d* as many times as is necessary until the signals received by the fourth signal beam **204***d* satisfy the stray tag threshold, as described herein.

**[0081]** Additionally, in certain embodiments, the reader **202** may iteratively adjust the orientation/direction of a signal beam until the reader **202** determines that the signal beam should be deactivated. The reader **202** may receive data from a particular signal beam that fails to satisfy the stray tag threshold, and the reader **202** may adjust the orientation/direction of the particular signal beam. However, when determining the orientation/direction adjustment, the reader **202** may also determine that the resulting direction or the resulting coverage area of the particular signal beam fails to satisfy a coverage threshold relative to another signal beam. The coverage threshold may generally represent (i) a distance metric, below which, a signal beam is considered too close to another signal beam, (ii) a coverage area metric, above which, a signal beam is considered too close to another signal beam, and/or any other suitable metric(s) or combinations thereof. Regardless, when the reader **202** determines the coverage threshold is violated, the reader **202** may deactivate the signal beam to be adjusted or the non-adjusted signal beam.

**[0082]** As an example, assume that the reader **202** determines the orientation/direction of the first signal beam **204***a* should be adjusted, such that the first signal beam **204***a* would move substantially closer to the second signal beam **204***b*. After the adjustment, the new orientation/direction of the first signal beam **204***a* would be zero degrees in the azimuthal direction and 33 degrees in the elevation direction. The reader **202** (e.g., via the beam adjustment algorithm **102***b***1**) may compare the new orientation/direction of the first signal beam **204***a* to the current orientation/direction of the second signal beam **204***b* and determine that the first signal beam **204***a* would have significant coverage area overlap with the second signal beam **204***b*. In fact, the reader **202** may determine that the distance between the center of the first signal beam **204***a* and the center of the second signal beam **204***b* (e.g., 3 degrees in elevation) may be small enough to violate the coverage threshold, and/or that the coverage area overlap between the first signal beam **204***a* and the second signal beam **204***b* may be large enough to violate the coverage threshold. In either case, the reader **202** may determine that either the first signal beam **204***a* or the second signal beam **204***b* should be deactivated to reduce tag read redundancy, and may deactivate one of the signal beams **204***a*, **204***b*.

**[0083]** FIG. **3**A depicts a first example asset tracking scenario **300** including the signal beam configuration **200** of FIG. **2**A, where the first signal beam **204***a* is adjusted to reduce the number of stray tags within the first signal beam **204***a* coverage area, in accordance with various embodiments described herein. More specifically, the first example asset tracking scenario **300** is a dock door truck loading/unloading scenario. In a typical dock door truck loading scenario, moving tags become stationary and there is an accumulation of layers or "walls" of stationary (i.e., stray) tags, such as the outer layer **302***a* and the inner layer **302***b*. Further, in either a loading scenario or an unloading scenario, the inner layer **302***b* is built first or removed last, such that the inner layer **302***b* comprises stray tags longer than any other layer.

**[0084]** Directing a signal beam (e.g., first signal beam **204***a*) toward the inner layer **302***b* may thus needlessly saturate the signals received by the signal beam **204***a* with stray tag reads. To avoid these issues, the reader **202** may adjust the orientation/direction of the first signal beam **204***a*, as illustrated by the adjustment **306**. Thus, as assets **304** are transported to/from the tag wall, the reader **202** may periodically adjust the orientation/direction of the first signal beam **204***a* to avoid placing the first signal beam **204***a* coverage area over the most forward tag wall layer (e.g., outer layer **302***a*). Additionally, or alternatively, the reader **202** may adjust the orientation/direction of the first signal beam **204***a* such that only the outer layer **302***a* of tags are monitored for wall break down (i.e., unloading) or buildup (i.e., loading). In this manner, the reader **202** may reduce the number of stationary tags monitored by any individual signal beam and thereby improve the detection accuracy of dynamic tag movement.

**[0085]** FIG. **3**B depicts a second example asset tracking scenario **310** including the signal beam configuration **200** of FIG. **2**A, where one or more signal beams (e.g., first signal beam **204***a* and/or fifth signal beam **204***e*) are adjusted to reduce the number of stray tags within the signal beam coverage area, in accordance with various embodiments described herein. More specifically, the second example asset tracking scenario **310** is a retail store entry/exit portal scenario, where the reader **202** may track the movement directions of tagged items across the entry/exit portal. In such scenarios, stray tags or tags that become stationary are common, and the beam adjustment performed by the reader **202** through the beam adjustment algorithm **102***b***1** may help focus the signal beams **204***a-f* on dynamic tags.

**[0086]** As illustrated in FIG. **3**B, the fifth signal beam **204***e* may originally be oriented at least partially into the third zone **206***c*, where multiple stationary assets **312***a* and corresponding stray tags are located. To reduce the impact of these stationary assets **312***a*, the reader **202** may adjust the orientation/direction of the fifth signal beam **204***e*, as illustrated by the adjustment **312***c*. As a result of this adjustment **312***c*, the fifth signal beam **204***e* may more frequently capture data from dynamic tags **312***b* that are entering/leaving the retail environment through the entry/exit portal.

**[0087]** Similarly, the first signal beam **204***a* may originally be oriented far enough outside of the entry/exit portal that the first signal beam **204***a* is unable to capture data from many dynamic tags **314***a*, **314***b* leaving and/or entering the retail location. To increase the likelihood that the first signal beam **204***a* may capture dynamic tag data, the reader **202** may adjust the orientation/direction of the first signal beam **204***a*, as illustrated by the adjustment **314***c*. As a result of this adjustment **314***c*, the first signal beam **204***a* may more frequently capture data from dynamic tags **314***a*, **314***b* that are entering/leaving the retail environment through the entry/exit portal. However, in certain embodiments, the reader **202** may not adjust the orientation/direction of the first signal beam **204***a* if, for example, the first signal beam **204***a* does not capture data from enough stray tags to violate the stray tag threshold. In other words, in these embodiments, the reader **202** may not adjust the orientation/direction of the signal beams when the stray tag threshold is not exceeded/violated.

**[0088]** FIG. **4** is a flowchart representative of a method **400** for automated beam adjustment for portal directionality, in accordance with embodiments described herein. Generally, and as described herein, the method **400** for automated beam adjustment for portal directionality may cause the server **110**, reader **102**, and/or any tags (e.g., tags **106***a*, **107***a*, **108***a*) to determine adjustments to signal beam orientation/direction by evaluating a number of stray tags indicated in the data captured by a signal beam, as compared to a stray tag threshold. More specifically, the method **400** enables the server **110**, the reader **102**, and the tags (e.g., tags **106***a*, **107***a*, **108***a*) to enhance/improve the accuracy of dynamic tag/asset tracking by adjusting the direction/orientation of a signal beam based on the number of stray tags indicated in the data captured by such signal beam, as described herein. It is to be understood that any of the steps of the method **400** may be performed by, for example, the server **110**, the reader **102**, tags (e.g., tags **106***a*, **107***a*, **108***a*), and/or any other suitable components or combinations thereof discussed herein.

**[0089]** At block **402**, the method **400** includes transmitting, by a transceiver configured to emit a plurality of signal beams that each have a respective coverage area and are oriented in a respective direction, a first signal beam of the plurality of signal beams to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction. The method **400** further includes determining, by a beam adjustment algorithm, that a first tag threshold is violated based on a number of first tags located within the first respective coverage area (block **404**). The method **400** further includes adjusting, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction (block **406**).

**[0090]** Further, at block **408**, the method **400** includes determining, by the beam adjustment algorithm, that the first tag threshold is violated based on a subsequent number of first tags located within the first respective coverage area. The method **400** further includes iteratively adjusting, by the beam adjustment algorithm, the orientation of the first signal beam to a subsequent direction until the first tag threshold is satisfied (block **410**).

**[0091]** In some embodiments, a second number of first tags located within the first respective coverage area of the first signal beam while oriented in the second direction is less than the number of first tags located within the first respective coverage area of the first signal beam while oriented in the first direction.

**[0092]** In certain embodiments, the first direction and the second direction include a respective azimuth component and a respective elevation component, and the method **400** further includes adjusting the orientation of the first signal beam by: adjusting a first respective azimuth component of the first direction to a second respective azimuth component of the second direction; or adjusting a first respective elevation component of the first direction to a second respective elevation component of the second direction.

**[0093]** In some embodiments, the first tags correspond to stray tags, a second number of dynamic tags are also located within the first respective coverage area while the first signal beam is oriented in the first direction, and the first tag threshold corresponds to a ratio of stray tags to dynamic tags.

**[0094]** In certain embodiments, the method **400** further includes determining, by the beam adjustment algorithm, that (i) the second direction or (ii) the first respective coverage area of the first signal beam fails to satisfy a coverage threshold relative to a second signal beam of the plurality of signal beams; and deactivating the first signal beam.

**[0095]** In some embodiments, the first tags correspond to stray tags, and the method **400** further includes receiving, by the transceiver, a response signal from a respective tag at a first time indicating that the respective tag is located within the first respective coverage area; receiving, by the transceiver, a subsequent response signal from the respective tag at a second time that is different from the first time indicating that the respective tag is located within the first respective coverage area; and determining, by the beam adjustment algorithm, that the respective tag is a stray tag.

**[0096]** In certain embodiments, the method **400** further includes receiving, by the transceiver, a second response signal from a second respective tag at the first time indicating that the second respective tag is located within the first respective coverage area; receiving, by the transceiver, a third response signal from the second respective tag at the second time indicating that the respective tag is located within a second respective coverage area of a second signal beam of the plurality of signal beams; and determining, by the beam adjustment algorithm, that the second respective tag is an dynamic tag.

**[0097]** Of course, it is to be appreciated that the actions of the method **400** may be performed in any suitable order and any suitable number of times.

**[0098]** FIG. **5** is a block diagram representative of an example logic circuit capable of implementing example methods and/or operations described herein. As an example, the example logic circuit may be capable of implementing one or more components of the reader **102** of FIG. **1**. Of course, it should be understood that the example logic circuit may also include and/or otherwise access instructions and/or components of other components represented in FIG. **1** and/or elsewhere herein, such as the server **110**, the tags (e.g., tags **106***a*, **107***a*, **108***a*), and the like.

**[0099]** The example logic circuit of FIG. **5** is a processing platform **510** capable of executing instructions to, for example, implement operations of the example methods described herein, as may be represented by the flowcharts of the drawings that accompany this description. Other example logic circuits capable of, for example, implementing operations of the example methods described herein include field programmable gate arrays (FPGAs) and application specific integrated circuits (ASICs).

**[0100]** The example processing platform **510** of FIG. **5** includes a processor **511** such as, for example, one or more microprocessors, controllers, and/or any suitable type of processor. The example processing platform **510** of FIG. **5** includes memory (e.g., volatile memory, non-volatile memory) **102***b* accessible by the processor **511** (e.g., via a memory controller). The example processor **511** interacts with the memory **102***b* to obtain, for example, machine-readable instructions stored in the memory **102***b* corresponding to, for example, the operations represented by the flowcharts of this disclosure. The memory **102***b* also includes the beam adjustment algorithm **102*b*1** that are accessible by the example processor **511**.

**[0101]** The beam adjustment algorithm **102*b*1** may comprise rule-based instructions configured to, for example, cause the example processor **511** to analyze data received from tags to determine adjustments and/or adjust orientations/directions of signal beams emitted by the reader **102**. More specifically, the beam adjustment algorithm **102*b*1** may comprise rule-based instructions configured to cause the example processor **511** to analyze data packets and/or information extracted, captured, and/or otherwise derived from the data packets to determine a number of stray tags within the coverage area of any signal beam emitted by the reader **102**. The beam adjustment algorithm **102*b*1** may further comprise rule-based instructions configured to, for example, cause the example processor **511** to determine whether the number of stray tags captured by a particular signal beam exceeds and/or otherwise fails to satisfy the stray tag threshold and to determine and cause an adjustment to the direction/orientation of the particular signal beam that reduces the number of captured stray tags.

**[0102]** To illustrate, the example processor **511** may access the memory **102*b*** to execute, reference, and/or otherwise interpret the beam adjustment algorithm **102*b*1** when receiving data packets from a proximate tag. Additionally, or alternatively, machine-readable instructions corresponding to the example operations described herein may be stored on one or more removable media (e.g., a compact disc, a digital versatile disc, removable flash memory, etc.) that may be coupled to the processing platform **510** to provide access to the machine-readable instructions stored thereon.

**[0103]** The example processing platform **510** of FIG. **5** also includes a radio transceiver **513** to enable communication with other machines via, for example, one or more networks. The example radio transceiver **513** includes any suitable type of communication interface(s) (e.g., wired and/or wireless interfaces) configured to operate in accordance with any suitable protocol(s) (e.g., Ethernet for wired communications, and/or BLE or IEEE 802.11 for wireless communications).

**[0104]** The example processing platform **510** of FIG. **5** also includes input/output (I/O) interfaces **512** to enable receipt of user input and communication of output data to the user. Such user input and communication may include, for example, any number of keyboards, mice, USB drives, optical drives, screens, touchscreens, etc.

**[0105]** Further, the example processing platform **510** may be connected to a remote server **520**. The remote server **520** may include one or more remote processors **522**, and may be configured to execute instructions to, for example, implement operations of the example methods described herein, as may be represented by the flowcharts of the drawings that accompany this description.

## ADDITIONAL CONSIDERATIONS

**[0106]** The above description refers to a block diagram of the accompanying drawings. Alternative implementations of the example represented by the block diagram includes one or more additional or alternative elements, processes and/or devices. Additionally, or alternatively, one or more of the example blocks of the diagram may be combined, divided, re-arranged or omitted. Components represented by the blocks of the diagram are implemented by hardware, software, firmware, and/or any combination of hardware, software and/or firmware. In some examples, at least one of the components represented by the blocks is implemented by a logic circuit. As used herein, the term "logic circuit" is expressly defined as a physical device including at least one hardware component configured (e.g., via operation in accordance with a predetermined configuration and/or via execution of stored machine-readable instructions) to control one or more machines and/or perform operations of one or more machines. Examples of a logic circuit include one or more processors, one or more coprocessors, one or more microprocessors, one or more controllers, one or more digital signal processors (DSPs), one or more application specific integrated circuits (ASICs), one or more field programmable gate arrays (FPGAs), one or more microcontroller units (MCUs), one or more hardware accelerators, one or more special-purpose computer chips, and one or more system-on-a-chip (SoC) devices. Some example logic circuits, such as ASICs or FPGAs, are specifically configured hardware for performing operations (e.g., one or more of the operations described herein and represented by the flowcharts of this disclosure, if such are present). Some example logic circuits are hardware that executes machine-readable instructions to perform operations (e.g., one or more of the operations described herein and represented by the flowcharts of this disclosure, if such are present). Some example logic circuits include a combination of specifically configured hardware and hardware that executes machine-readable instructions. The above description refers to various operations described herein and flowcharts that may be appended hereto to illustrate the flow of those operations. Any such flowcharts are representative of example methods disclosed herein. In some examples, the methods represented by the flowcharts implement the apparatus represented by the block diagrams. Alternative implementations of example methods disclosed herein may include additional or alternative operations. Further, operations of alternative implementations of the methods disclosed herein may combined, divided, re-arranged or omitted. In some examples, the operations described herein are implemented by machine-readable instructions (e.g., software and/or firmware) stored on a medium (e.g., a tangible machine-readable medium) for execution by one or more logic circuits (e.g., processor(s)). In some examples, the operations described herein are implemented by one or more configurations of one or more specifically designed logic circuits (e.g., ASIC (s)). In some examples the operations described herein are implemented by a combination of specifically designed logic circuit(s) and machine-readable instructions stored on a medium (e.g., a tangible machine-readable medium) for execution by logic circuit(s).

**[0107]** As used herein, each of the terms "tangible machine-readable medium," "non-transitory machine-readable medium" and "machine-readable storage device" is expressly defined as a storage medium (e.g., a platter of a hard disk drive, a digital versatile disc, a compact disc, flash memory, read-only memory, random-access memory, etc.) on which machine-readable instructions (e.g., program code in the form of, for example, software and/or firmware) are stored for any suitable duration of time (e.g., permanently, for an extended period of time (e.g., while a program associated with the machine-readable instructions is executing), and/or a short period of time (e.g., while the machine-readable instructions are cached and/or during a buffering process)). Further, as used herein, each of the terms "tangible machine-readable medium," "non-transitory machine-readable medium" and "machine-readable storage device" is expressly defined to exclude propagating signals. That is, as used in any claim of this patent, none of the terms "tangible machine-readable medium," "non-transitory machine-readable medium," and "machine-readable storage device" can be read to be implemented by a propagating signal.

**[0108]** In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings. Additionally, the described embodiments/examples/implementations should not be interpreted as mutually exclusive, and should instead be understood as potentially combinable if such combinations are permissive in any way. In other words, any feature disclosed in any of the aforementioned embodiments/examples/implementations may be included in any of the other aforementioned embodiments/examples/implementations.

**[0109]** The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The claimed invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

**[0110]** Moreover, in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms "comprises," "comprising," "has", "having," "includes", "including," "contains", "containing" or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element proceeded by "comprises . . . a", "has . . . a", "includes . . . a", "contains . . . a" does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms "a" and "an" are defined as one or more unless explicitly stated otherwise herein. The terms "substantially", "essentially", "approximately", "about" or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term "coupled" as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is "configured" in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

**[0111]** The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may lie in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

**1**. An assembly comprising:

a transceiver configured to emit a plurality of signal beams that each have a respective coverage area and are oriented in a respective direction;

one or more processors; and

one or more memories communicatively coupled to the one or more processors storing a beam adjustment algorithm and instructions that, when executed by the one or more processors, cause the assembly to:

transmit, by the transceiver, a first signal beam of the plurality of signal beams to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction,

determine, by the beam adjustment algorithm, that a first tag threshold is violated based on a number of first tags located within the first respective coverage area, and

adjust, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction, wherein

the first tags correspond to stray tags, a second number of dynamic tags are also located within the first respective coverage area while the first signal beam is oriented in the first direction, and the first tag threshold corresponds to a ratio of stray tags to dynamic tags.

**2**. The assembly of claim **1**, wherein a second number of first tags located within the first respective coverage area of the first signal beam while oriented in the second direction is less than the number of first tags located within the first respective coverage area of the first signal beam while oriented in the first direction.

**3**. The assembly of claim **1**, wherein the first direction and the second direction include a respective azimuth component and a respective elevation component, and the instructions, when executed by the one or more processors, further cause the assembly to adjust the orientation of the first signal beam by:

adjusting a first respective azimuth component of the first direction to a second respective azimuth component of the second direction; or

adjusting a first respective elevation component of the first direction to a second respective elevation component of the second direction.

**4**. (canceled)

**5**. The assembly of claim **1**, wherein after adjusting the orientation of the first signal beam to the second direction, the instructions, when executed by the one or more processors, further cause the assembly to:

(a) determine, by the beam adjustment algorithm, that the first tag threshold is violated based on a subsequent number of first tags located within the first respective coverage area;

(b) adjust, by the beam adjustment algorithm, the orientation of the first signal beam to a subsequent direction; and

iteratively perform steps (a) and (b) until the first tag threshold is satisfied.

**6**. The assembly of claim **1**, wherein the instructions, when executed by the one or more processors, further cause the assembly to:

determine, by the beam adjustment algorithm, that (i) the second direction or (ii) the first respective coverage area of the first signal beam fails to satisfy a coverage threshold relative to a second signal beam of the plurality of signal beams; and

deactivate the first signal beam.

**7**. The assembly of claim **1**, wherein first tags correspond to stray tags, and the instructions, when executed by the one or more processors, further cause the assembly to:

receive, by the transceiver, a response signal from a respective tag at a first time indicating that the respective tag is located within the first respective coverage area;

receive, by the transceiver, a subsequent response signal from the respective tag at a second time that is different from the first time indicating that the respective tag is located within the first respective coverage area; and

determine, by the beam adjustment algorithm, that the respective tag is a stray tag.

**8**. The assembly of claim **7**, wherein the instructions, when executed by the one or more processors, further cause the assembly to:

receive, by the transceiver, a second response signal from a second respective tag at the first time indicating that the second respective tag is located within the first respective coverage area;

receive, by the transceiver, a third response signal from the second respective tag at the second time indicating that the respective tag is located within a second respective coverage area of a second signal beam of the plurality of signal beams; and

determine, by the beam adjustment algorithm, that the second respective tag is an dynamic tag.

**9**. A method comprising:

transmitting, by a transceiver configured to emit a plurality of signal beams that each have a respective coverage area and are oriented in a respective direction, a first signal beam of the plurality of signal beams to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction;

determining, by a beam adjustment algorithm, that a first tag threshold is violated based on a number of first tags located within the first respective coverage area; and

adjusting, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction, wherein

the first tags correspond to stray tags, a second number of dynamic tags are also located within the first respective coverage area while the first signal beam is oriented in the first direction, and the first tag threshold corresponds to a ratio of stray tags to dynamic tags.

**10**. The method of claim **9**, wherein a second number of first tags located within the first respective coverage area of the first signal beam while oriented in the second direction is less than the number of first tags located within the first respective coverage area of the first signal beam while oriented in the first direction.

**11**. The method of claim **9**, wherein the first direction and the second direction include a respective azimuth component and a respective elevation component, and the method further comprises adjusting the orientation of the first signal beam by:

adjusting a first respective azimuth component of the first direction to a second respective azimuth component of the second direction; or

adjusting a first respective elevation component of the first direction to a second respective elevation component of the second direction.

**12**. (canceled)

**13**. The method of claim **9**, wherein after adjusting the orientation of the first signal beam to the second direction, the method further comprises:

(a) determining, by the beam adjustment algorithm, that the first tag threshold is violated based on a subsequent number of first tags located within the first respective coverage area;

(b) adjusting, by the beam adjustment algorithm, the orientation of the first signal beam to a subsequent direction; and

iteratively performing steps (a) and (b) until the first tag threshold is satisfied.

**14**. The method of claim **9**, further comprising:

determining, by the beam adjustment algorithm, that (i) the second direction or (ii) the first respective coverage area of the first signal beam fails to satisfy a coverage threshold relative to a second signal beam of the plurality of signal beams; and

deactivating the first signal beam.

**15**. The method of claim **9**, wherein first tags correspond to stray tags, and the method further comprises:

receiving, by the transceiver, a response signal from a respective tag at a first time indicating that the respective tag is located within the first respective coverage area;

receiving, by the transceiver, a subsequent response signal from the respective tag at a second time that is different from the first time indicating that the respective tag is located within the first respective coverage area; and

determining, by the beam adjustment algorithm, that the respective tag is a stray tag.

**16**. The method of claim **15**, further comprising:

receiving, by the transceiver, a second response signal from a second respective tag at the first time indicating that the second respective tag is located within the first respective coverage area;

receiving, by the transceiver, a third response signal from the second respective tag at the second time indicating that the respective tag is located within a second respective coverage area of a second signal beam of the plurality of signal beams; and

determining, by the beam adjustment algorithm, that the second respective tag is an dynamic tag.

**17**. A tangible machine-readable medium comprising instructions that, when executed, cause a machine to at least:

transmit, by a transceiver configured to emit a plurality of signal beams that each have a respective coverage area and are oriented in a respective direction, a first signal beam of the plurality of signal beams to a set of tags located within a first respective coverage area of the first signal beam while the first signal beam is oriented in a first direction;

determine, by a beam adjustment algorithm, that a first tag threshold is violated based on a number of first tags located within the first respective coverage area; and

adjust, by the beam adjustment algorithm, an orientation of the first signal beam from the first direction to a second direction, wherein

the first tags correspond to stray tags, a second number of dynamic tags are also located within the first respective coverage area while the first signal beam is oriented in the first direction, and the first tag threshold corresponds to a ratio of stray tags to dynamic tags.

**18**. The tangible machine-readable medium of claim **17**, wherein the first direction and the second direction include a respective azimuth component and a respective elevation component, and the instructions, when executed, further cause the machine to adjust the orientation of the first signal beam by:

adjusting a first respective azimuth component of the first direction to a second respective azimuth component of the second direction; or

adjusting a first respective elevation component of the first direction to a second respective elevation component of the second direction.

**19**. (canceled)

**20**. The tangible machine-readable medium of claim **17**, wherein after adjusting the orientation of the first signal beam to the second direction, the instructions, when executed, further cause the machine to:

(a) determine, by the beam adjustment algorithm, that the first tag threshold is violated based on a subsequent number of first tags located within the first respective coverage area;

(b) adjust, by the beam adjustment algorithm, the orientation of the first signal beam to a subsequent direction; and

iteratively perform steps (a) and (b) until the first tag threshold is satisfied.

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